

The Metro Is the Market: Why Local Airbnb Regulations May Not Lower Rents

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October 29, 2026

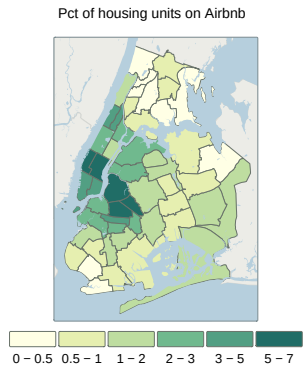
Background

- Since its launch in 2008, Airbnb has rapidly grown to become a major part of the tourism industry
- Many cities have restricted short-term rentals (STRs) to improve housing affordability
- Intuition: STRs use housing which could otherwise be rented to residents, so restricting STRs should increase long-term rental (LTR) supply and lower rents

Airbnb in New York City

In January 2023, NYC had approximately 38,500 active non-hotel Airbnb listings

- 15-20% of total hotel stock in NYC, highest share in the US (Farronato and Fradkin, 2022)
- Over 2% of total rental stock in NYC
- Roughly matched the number of vacant rental units



NYC's Local Law 18

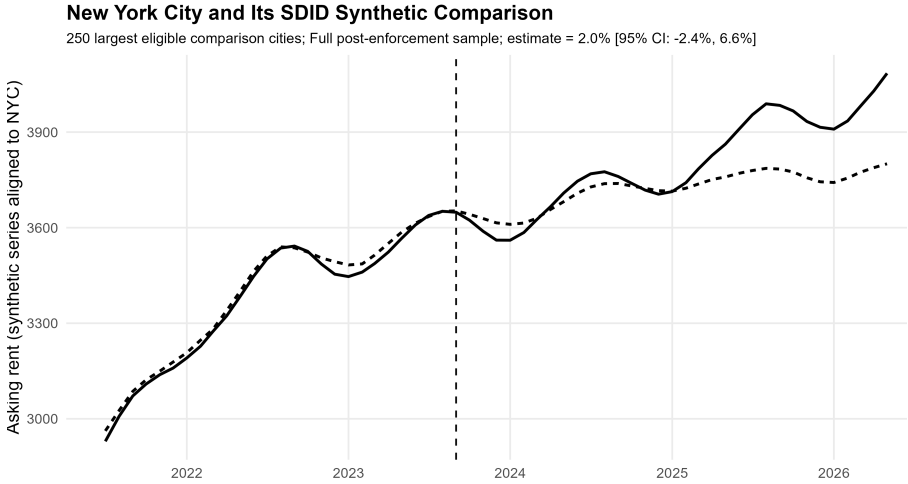
In September 2023, NYC began enforcing the strictest STR regulations in the US

- Hosts must register and be physically present during stays
- Limited to two guests per stay
- Platforms prohibited from processing unregistered bookings
- Airbnb listings fell by 90% overnight

Should have caused significant rent decrease, particularly in Manhattan

Aftermath of Local Law 18

Marsella (2026) estimates a rent effect in first 12 months of only -0.1%.



This Paper

In this paper I provide a simple model of a city with residents and visitors that explains this puzzle

- Short-term rental regulation can return housing to the long-term rental market in one area of the city
- But displaced tourists spread out and find accommodation in other parts of the city
- The metropolitan area is the relevant market for housing
- If other areas of the city can absorb the displaced tourists, long-term rents may not fall anywhere

Testable Implications

Restricting short-term rentals in the model:

- Leaves long-term rents unchanged
- Raises short-term accommodation prices
- Shifts short-term rental activity away from the city centre

I evaluate these predictions using evidence from New York City's Local Law 18

As a result:

- Long-term resident welfare is unaffected or made worse off by regulation
- Short-term renter welfare declines

Contribution to the literature

Housing market effects of STR regulation

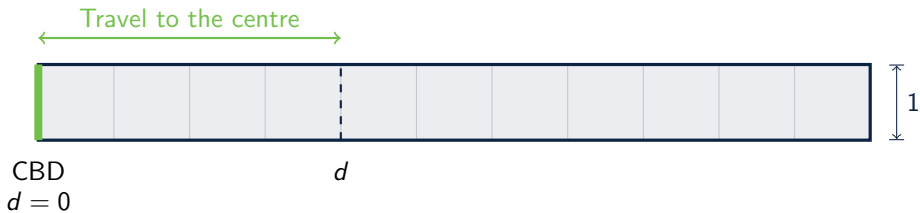
Calder-Wang (2022); Koster et al. (2021); Seiler et al. (2024); Duso et al. (2024); Bibler et al. (2025); van Holm (2020); Valentin (2021); Marsella (2026)

I develop a spatial equilibrium model explaining how local STR restrictions can raise short-term rents without lowering residential rents, and test its predictions using NYC's Local Law 18

▸ Affordability Estimates

Environment

City with jobs and tourist destinations at the CBD and land stretches out along a line



- Residents and visitors choose where to live or stay, trading off housing costs and distance from the centre
- Absentee landlords allocate housing between long-term and short-term rentals
- Regulation limits the share of housing available for short-term use

Residents

Two groups of fixed size choose locations:

- N_R residents, each requiring $\bar{\ell}_R$ of housing
- N_V visitors, each requiring $\bar{\ell}_V$

Each chooses distance d and numeraire consumption x to maximize $u(x)$ subject to $x = w_i - t_i d - r^i(d)\bar{l}_i$. Indifference across locations gives the **bid-rent function**

$$R^i(d) = \frac{w_i - t_i d - h(\bar{u}^i)}{\bar{\ell}_i}, \qquad \text{slope } -\frac{t_i}{\bar{\ell}_i}$$

Assumption 1: Visitors consume less housing and/or have higher transport costs ($t_V/\bar{\ell}_V > t_R/\bar{\ell}_R$), so their bid-rent gradient is steeper $\Rightarrow$ visitors outbid residents near the centre

Landlords

Each landlord faces an idiosyncratic net cost of short-term renting $\varepsilon \sim F$, with support on $[0, \infty]$ (turnover, management, building rules)

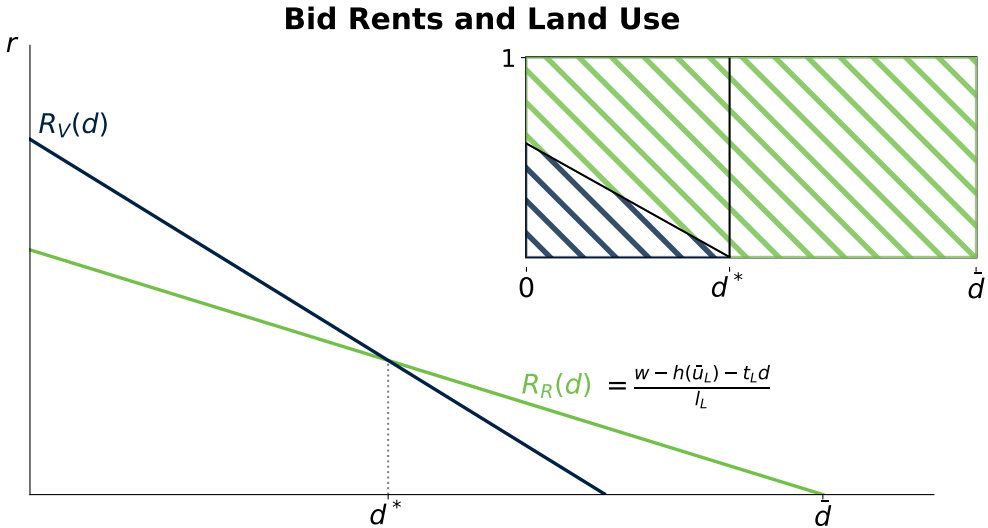
A landlord at d rents to a visitor if

$$R^V(d) - R^R(d) > \varepsilon$$

So the share of land in short-term use $\Gamma(d)$ is continuous and increasing in the bid-rent gap

- Near the centre the gap is large $\Rightarrow$ STRs concentrate downtown
- Beyond the boundary d^* where $R^V = R^R$, no landlord chooses STR

Initial Conditions



Equilibrium

Set of utility levels and bid rents $\{\bar{u}_V, \bar{u}_R\}$, bid rents $\{R^V(d), R^R(d)\}$, an allocation $\Gamma(d)$, and a city boundary $\bar{d}$ such that:

1. $R^i(d) = \frac{w_i - t_i d - h(\bar{u}^i)}{\bar{\ell}_i}$
2. $\Gamma(d) = F(R^V(d) - R^R(d)))$
3. $\int_0^{\bar{d}} \Gamma(d)\,\mathrm{d}d = N_V \bar{\ell}_V$
4. $\int_0^{\bar{d}} [1 - \Gamma(d)]\,\mathrm{d}d = N_R \bar{\ell}_R$
5. $R_R(\bar{d}) = R_A$

Policy

A fraction $\alpha \in [0, 1]$ of housing at each location is eligible for short-term rental use

Combining the regulation with the landlord's decision, the fraction of landlords at d who rent to visitors is

$$\Gamma(d, \alpha) \equiv \alpha F\Big(R^V(d) - R^R(d)\Big)$$

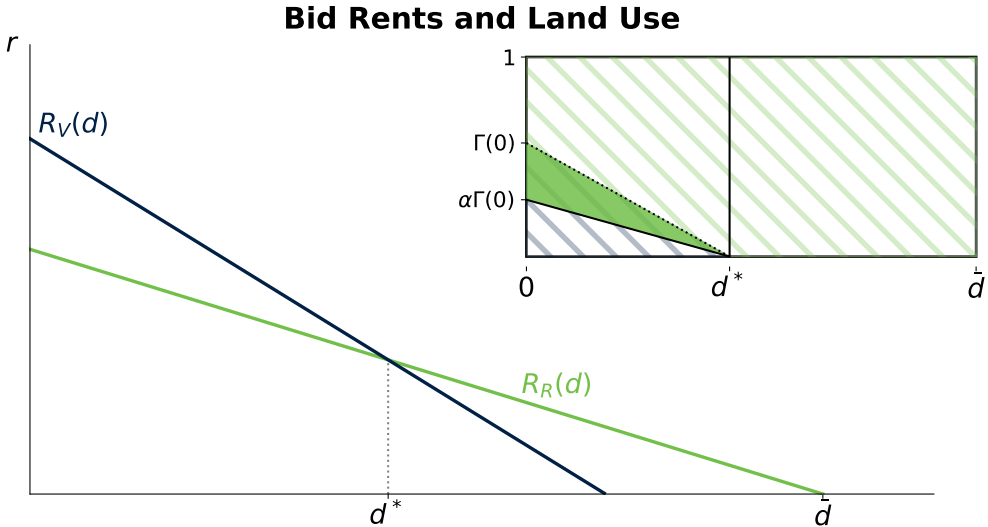
Equilibrium with policy

Set of utility levels $\{\bar{u}_V, \bar{u}_R\}$, bid rents $\{R^V(d), R^R(d)\}$, an allocation $\Gamma(d, \alpha)$, and a city boundary $\bar{d}$ such that:

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4. $\int_0^{\bar{d}} [1 - \Gamma(d, \alpha)]\, \mathrm{d}d = N_R \bar{\ell}_R$
5. $R_R(\bar{d}) = R_A$

Because $\bar{\ell}_V < \bar{\ell}_R$, visitors outbid residents near center $\Rightarrow$ STR concentrated at low d

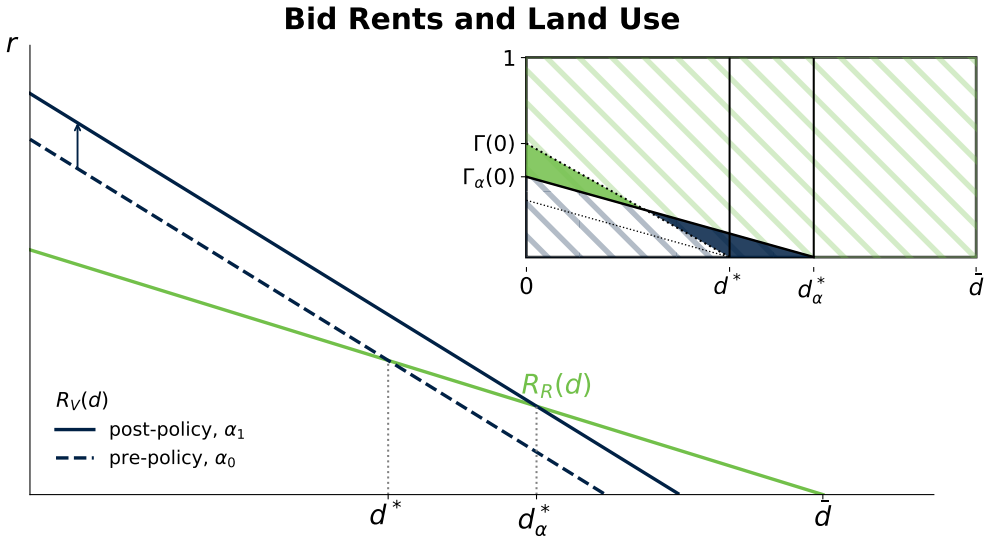
Regulation displaces tourists



Assumption 2: Number of tourists stays the same

- Assumption on N_V is critical for the model's predictions
- Visitors are choosing between destination cities based on a separate model with many determinants beyond Airbnb prices
- As a baseline, I assume that the number of visitors to the city is fixed
- Later I provide evidence that this assumption is reasonable for New York City during the study period

Tourists relocate



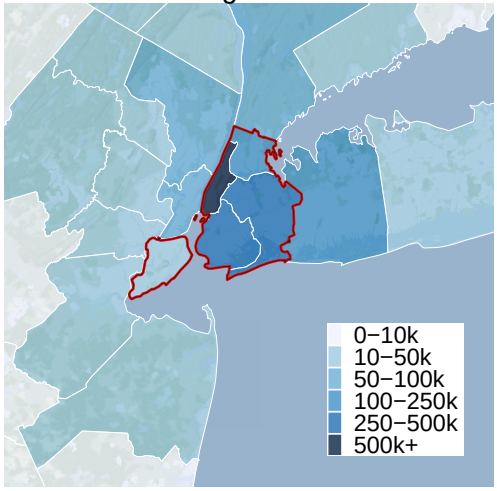
Propositions

- **Proposition 1:** STR regulation does not change long-term rents or resident welfare
- **Proposition 2:** STR regulation increases STR prices and decreases tourist welfare
- **Proposition 3:** STR regulation flattens the STR supply gradient

▸ Decomposition

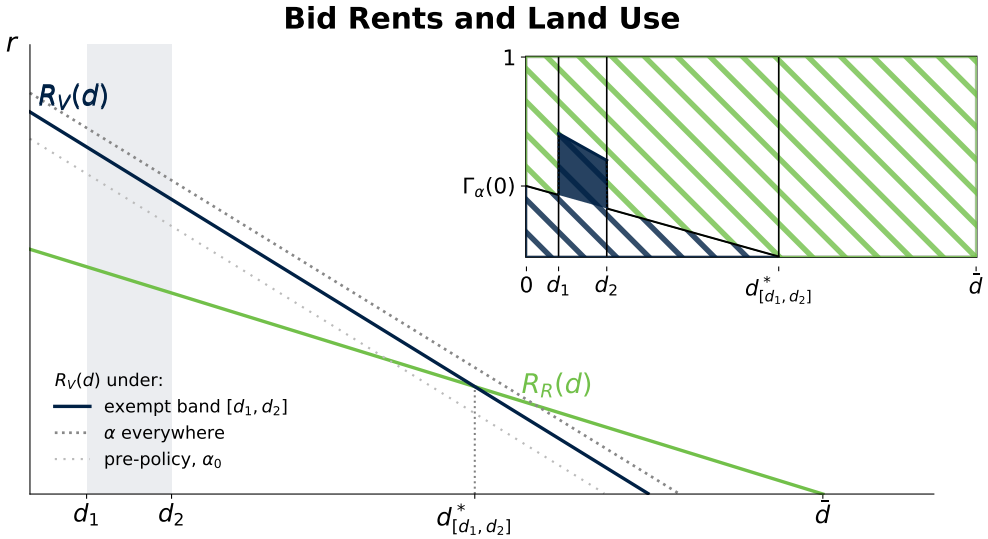
NYC Setting

Residents working in Manhattan



2016−2020 ACS.

Municipal boundary extension



Propositions

- **Proposition 1:** STR regulation does not change long-term rents or resident welfare
- **Proposition 2:** STR regulation increases STR prices and decreases tourist welfare
- **Proposition 3:** STR regulation flattens the STR supply gradient
- **Proposition 4:** Localized STR regulation causes a discontinuity of STR supply at the municipal border

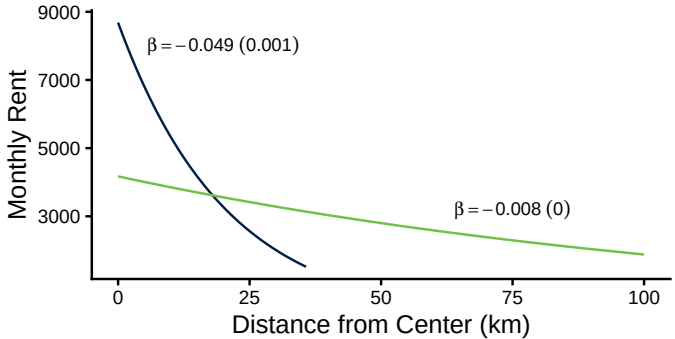
▸ Decomposition

Empirics

Validating Assumption 1

Assumption 1: Tourists have steeper bid-rent gradient than residents

$$\ln(\text{Monthly Rent}_{ij}) = \alpha_{ij} + \beta \text{Distance}_{ij} + \gamma X_{ij} + \varepsilon_{ij}$$



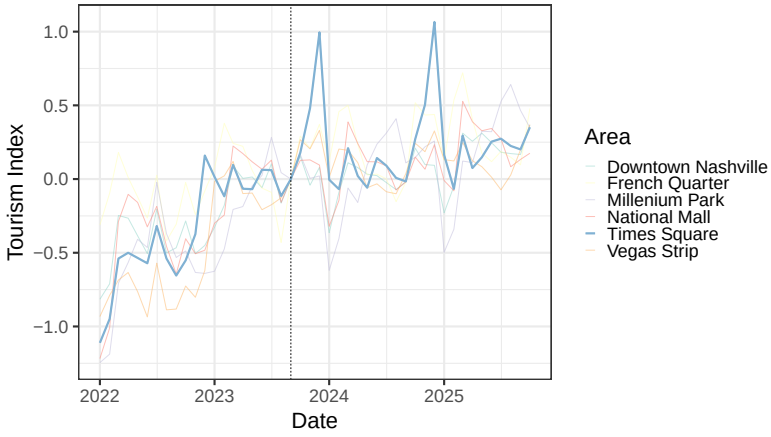
Source: InsideAirbnb and Dwellsy. Airbnb prices converted to monthly

Validating Assumption 2

Assumption 2: Number of tourists stays the same when regulation increases

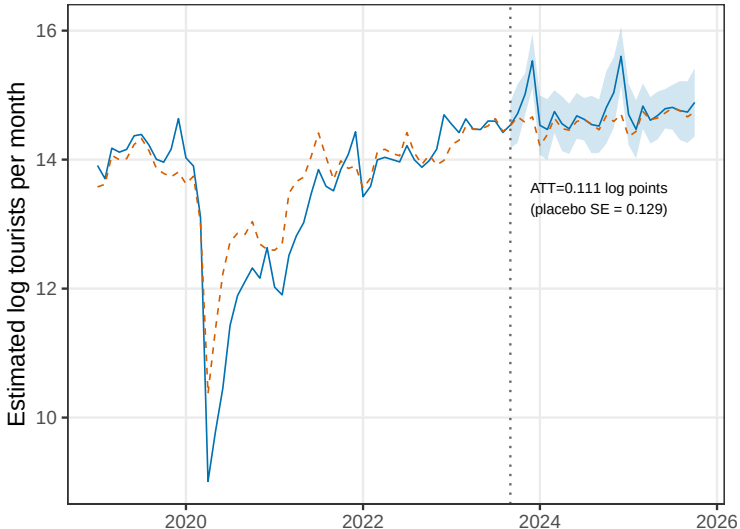
- Advan cell phone data, 2018-2025
- Problem: Tourism during study period was recovering from COVID-19 pandemic
- Compare tourist visits to Times Square with other major US tourist destinations before and after Local Law 18
  - e.g. French Quarter, New Orleans; Millenium Park, Chicago; Quincy Market, Boston; Independence Hall, Philadelphia; National Mall, Washington DC; Las Vegas Strip, Las Vegas
- Synthetic difference-in-differences

Tourist Visits to NYC did not Decrease



Source: Advan Neighborhood Patterns Plus.
Log tourist visits normalized to 0 in September 2023

Tourist Visits to NYC did not Decrease — Synthetic DiD

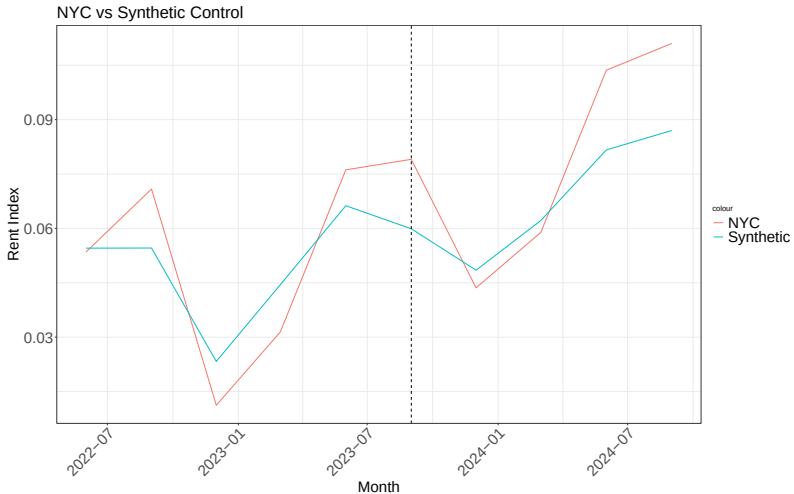


Rental Analysis

- **LTR:** Dwellsy (2021–25), 30% sample of US rental listings
  RentHub (2021–25), additional rental listings
- **Sales:** ATTOM (2021–25), universe of recorded sales in US

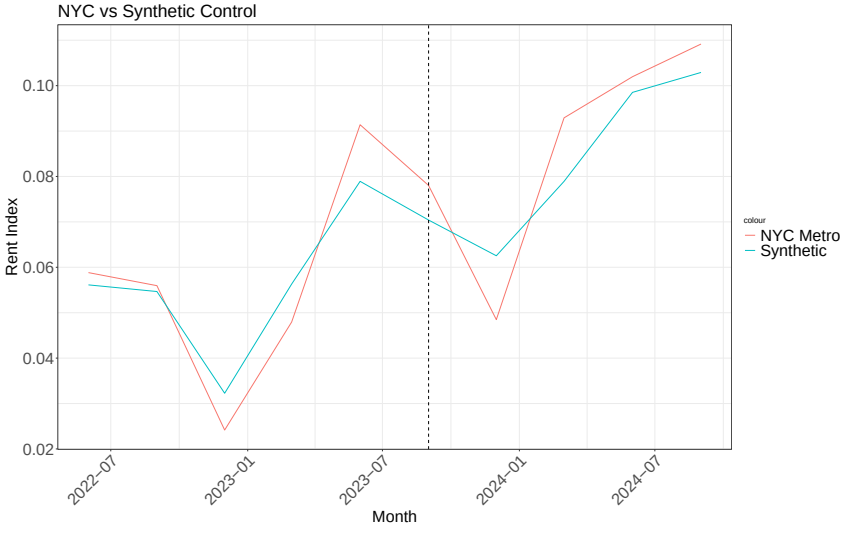
- **Synthetic DiD:** Synthetic control using US cities which did not implement major STR regulation during the time period

Proposition 1: LTR Rents — Synthetic DiD

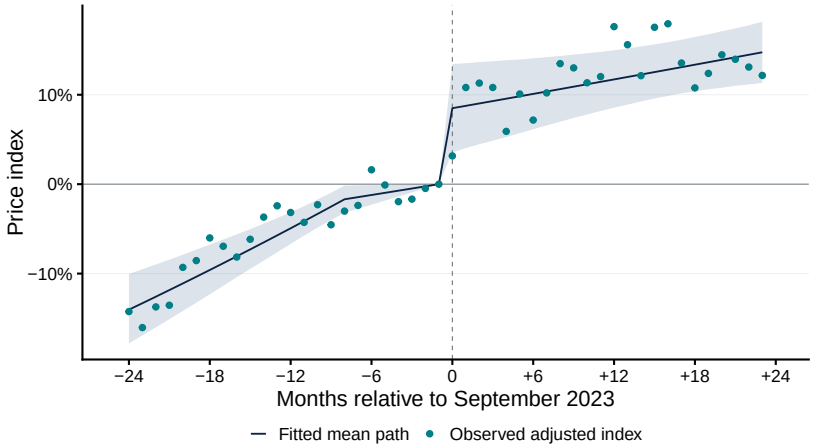


▸ Weights ▸ Full Sample Synthetic Control ▸ LA Replication

Proposition 1: LTR Rents — Metropolitan Area



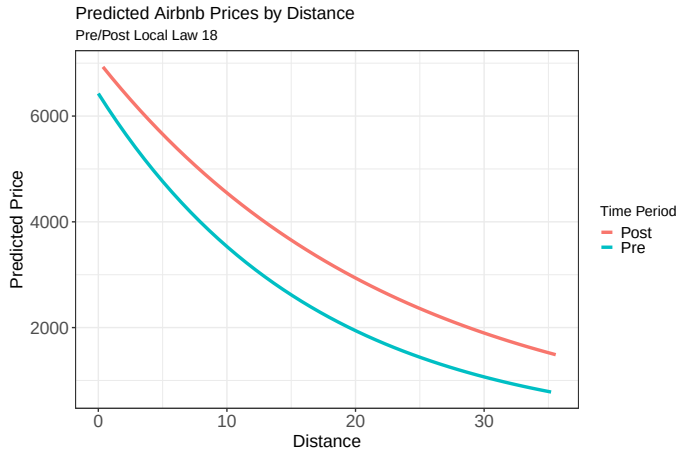
Proposition 2: STR Prices Increase At Regulation



Source: InsideAirbnb. Seasonally adjusted repeat−price index.
Normalized to 0 at August 2023.

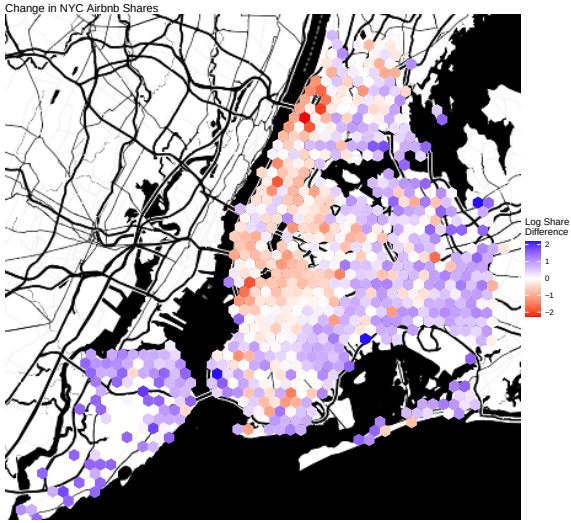
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Prediction: $R^S(d)$ shifts up uniformly



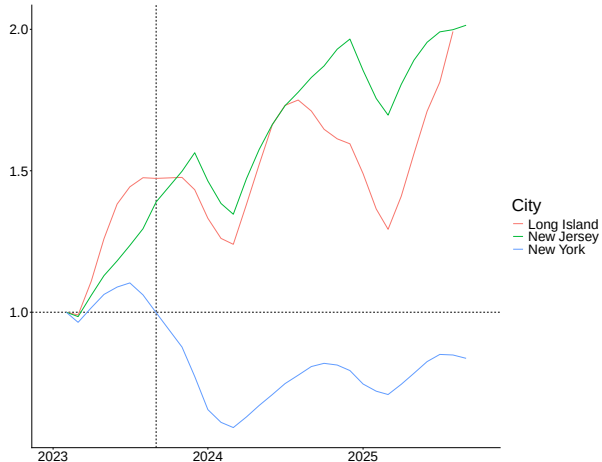
Proposition 3: STR Supply Gradient Flattens

Prediction: Within NYC, STR supply shifts from center to periphery



Proposition 4: STR Supply Shifts to Unregulated Areas

Prediction: STR supply decreases downtown, increases at periphery



January 2023 = 1. 3−month average of total listings in each city.
Source: AirDNA

Conclusion

Summary:

- Municipal STR regulation creates spatial arbitrage
- Tourists relocate; total LTR supply unchanged
- City loses tourist revenue without affordability gains
- Data from NYC supports model predictions

Policy implications:

- Place-based STR bans are ineffective for affordability
- Regional coordination needed to affect housing supply
- But policymakers should acknowledge affordability tradeoffs with tourism

Model: Discussion

Key assumption: Number of tourists stays the same when regulation increases

Rent decrease is possible if number of tourists decreases but then welfare is ambiguous

- Substitutability across destinations
- Strict enough regulations

▸ Other Models

Decomposition

$$\frac{dR_L}{d\alpha} = \underbrace{\left.\frac{\partial R_L}{\partial \alpha}\right|_{N_S}}_{=0} + \underbrace{\frac{\partial R_L}{\partial N_S}}_{\text{crowding}} \cdot \underbrace{\frac{dN_S}{dR_S}}_{\text{Airbnb demand elasticity}} \cdot \underbrace{\frac{dR_S}{d\alpha}}_{\text{Airbnb price shift}}$$

- **Spatial arbitrage:** Holding N_S fixed, regulation shifts STR supply spatially but R_L is pinned at $\bar{d}$

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Regulation improves affordability **iff** tourists actually leave

When Does Regulation Improve Affordability?

Let $s \equiv \frac{N_S}{N_S+N_L}$ denote the STR share of the housing stock.

$$\frac{d \ln R_L}{d \ln \alpha} = \underbrace{\frac{\partial \ln R_L}{\partial s}}_{\substack{\text{Rent–STR share} \\ \text{semi-elasticity}}} \cdot s \cdot \underbrace{\varepsilon_D}_{\substack{\text{Tourist demand} \\ \text{elasticity}}} \cdot \underbrace{\frac{d \ln R_S}{d \ln \alpha}}_{\substack{\text{Price–regulation} \\ \text{elasticity}}}$$

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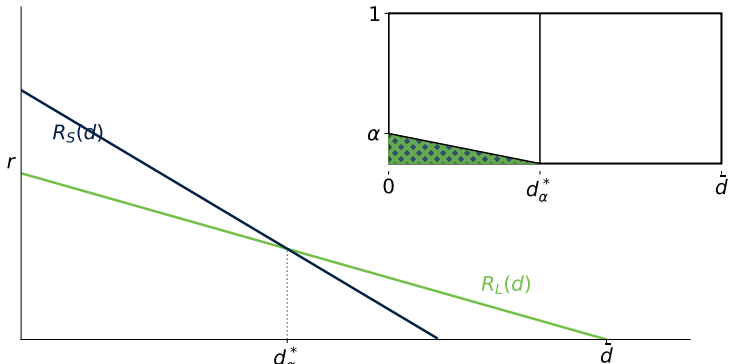
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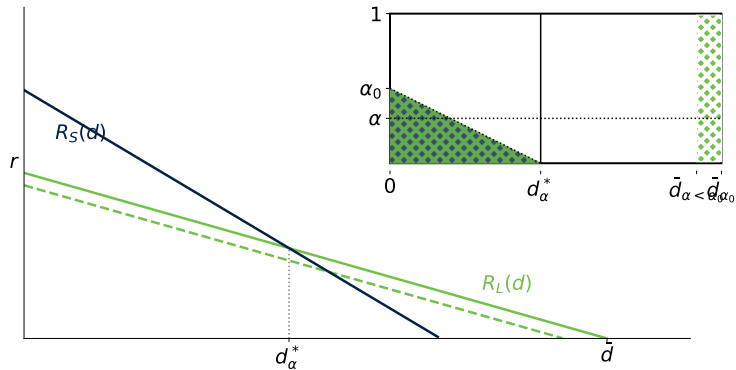
All four terms are small $\Rightarrow$ regulation has minimal affordability effect ◂ Back

How predictions change when we make different assumptions on N_S



- Tourists replaced by residents
- No change to bid rents or utility

Half Open



- Tourists replaced by residents so city shrinks
- Slight decrease in resident bid rent and increase in utility

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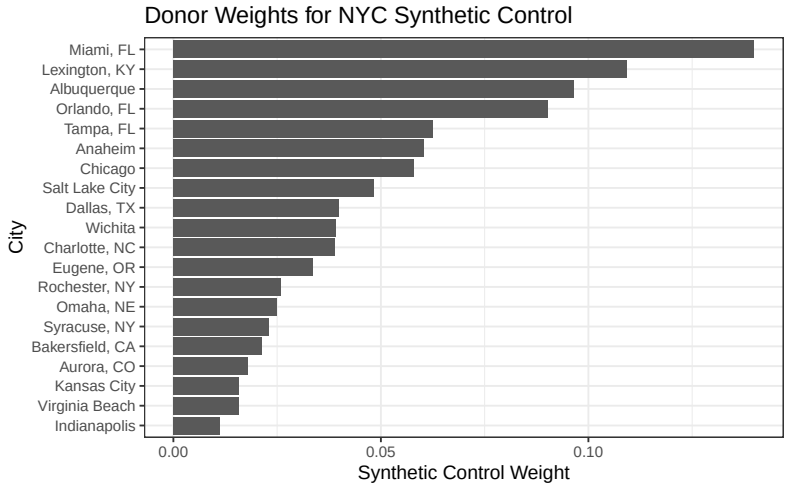
LTR Rents Table

	log(rent)	
	High Airbnb Zip Codes	NYC
	(1)	(2)
Post	0.0527***	0.0400***
	(0.003)	(0.0028)
Treated × Post	0.0059	0.0138***
	(0.0058)	(0.0038)
Observations	38,414	50,398
R^2	0.9215	0.9581
Within R^2	0.0232	0.0241
Property FE	✓	✓

Standard errors clustered at the property level.

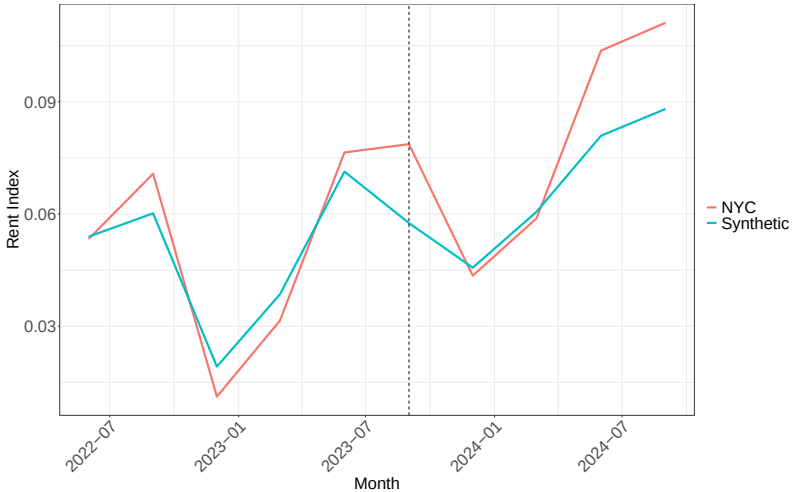
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Synthetic Control Weights



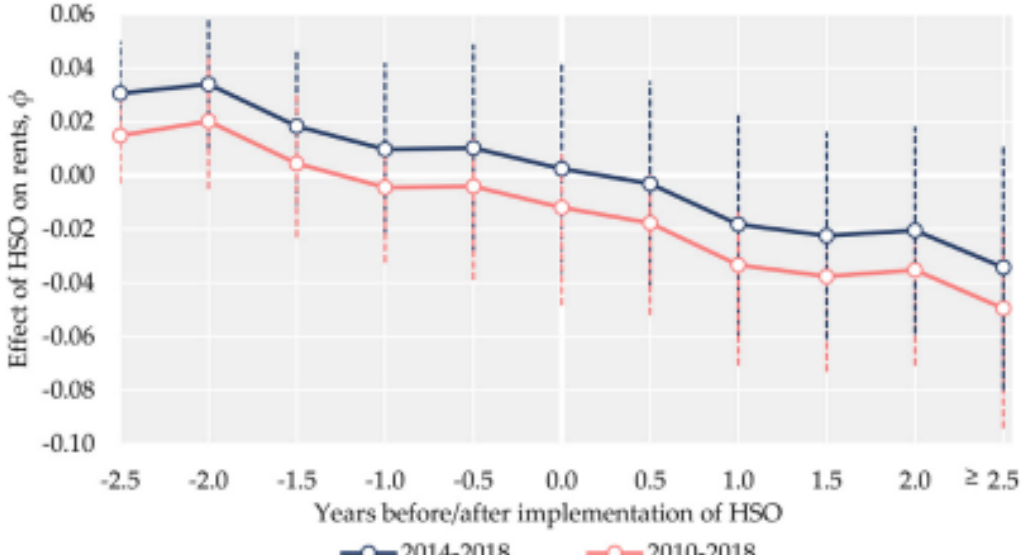
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Full Sample Synthetic Control



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Affordability Estimates Discrepancy



LA Rent Regulation — Staggered Synthetic DiD

Zillow AllHomes rents: Koster et al. (2021) setting, national ZIP donors.

Pre-treatment months	Treated ZIPs	Estimated rent change	
		Outside LA metro	Outside California
12	13	+0.38%	+0.44%
24	9	−0.18%	−0.06%
36	9	−0.64%	−0.40%

- Main donor pool excludes the LA–Long Beach–Anaheim metro; other California ZIPs remain eligible.
- Complete monthly histories; pre-2014 data allowed; follow-up through September 2018.
- Cohort estimates weighted by treated ZIP-months. The 12-month window retains more treated ZIPs.

Exploratory point estimates: donor policies unscreened; confidence intervals not yet calculated.

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